When Does Directional Reward Shaping Matter?

Same Dijkstra Regret, Two Learning Paradigms, One Graph-Walking Task

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Abstract

Potential-based shaping can preserve optimal policies, but that invariance result does not predict finite-budget learning speed or tell us whether an exact directional signal should enter as a reward or a supervised loss. We study shortest-path execution on connected weighted graphs under a strictly local observation. At each node, a policy sees at most the eight cheapest incident edges and must reach a hidden target. Dijkstra's potential gives the exact one-step regret $\Delta(u, v) = w(u, v) + \phi(v) - \phi(u)$. We expose the same quantity either as the reinforcement-learning reward $-\Delta/w_{\max}$ or as the weight in a teacher-forced soft-KL objective. Models are matched at two parameter scales and two estimated-compute budgets, trained with three seeds, and evaluated by both greedy and sampled decoding on three hash-pinned 1,000-graph suites.

At the 25M match, sampled RL attains 90.8% and 87.6% far-out-of-distribution (OOD) success at the small and large scales, versus 12.8% and 31.3% for supervision. RL also makes fewer backward moves and yields substantially cheaper successful walks. The predeclared cross-paradigm gate passes in 24 success cells, 24 optimality-gap cells, and six size-degradation comparisons. A separate 162-run shaping phase diagram produces a coherent 8–12 point advantage on large cold-start graphs with tight horizons, but no cell reaches its predeclared 15-point gate: an honest null. Finally, sampled MLP and LSTM policies retain 90.9% and 93.6% success at 24 times the largest training size, while a Transformer falls to 56.1%. High random-walk success and near-universal two-cycles show why reachability, efficiency, and decoding must be reported jointly. The results support a narrow conclusion: for this local graph walker, the learning interface and induced state distribution matter more than merely possessing an exact directional signal.

1 Introduction

Dijkstra's algorithm provides unusually clean supervision for testing learned graph search. If $\phi(x) = d(x, \text{target})$ is the exact shortest-path potential, every candidate move $u \to v$ has nonnegative regret

$$\Delta(u, v) = w(u, v) + \phi(v) - \phi(u) \geq 0, \tag{1}$$

with equality precisely for a shortest-path-consistent move. The same scalar can grade an expert action, shape an RL reward, or remain only an evaluation diagnostic. This lets us ask a controlled question obscured in most learned search comparisons: what changes when the information is held fixed but its learning interface changes?

Potential-based reward shaping (PBRS) can preserve an MDP's optimal policies under its standard discounted construction [9]. This theorem is silent about finite-budget optimization and does not equate teacher-forced learning with on-policy interaction. A supervised executor trains on expert states. An RL walker trains on states induced by its own errors, detours, and recoveries. Holding the local signal fixed isolates this difference more cleanly than comparing unrelated rewards and targets.

We make four contributions. First, we compare PPO-based RL and soft-KL supervision on identical frozen graph suites at two closely matched parameter scales and two approximate multiply–accumulate (MAC) budgets. Second, we predeclare and execute a reward-shaping phase diagram over graph size, horizon, and terminal bonus; its null is retained rather than tuned away. Third, we evaluate size generalization through $n = 768$, 24 times the largest training graph, with classical controls and cycle diagnostics. Fourth, we provide a deterministic seed path, hash-pinned suites, complete three-seed aggregates, and a behavior-preserving vectorized environment.

Our claims are deliberately narrower than "RL beats supervision" or "the network learns Dijkstra." They concern local-observation graph walkers, the exact regret in Equation (1), and a fixed family of connected random weighted graphs. Even in that controlled setting, success alone is misleading: a uniform random valid walk exceeds 80% success under a generous $4n$ horizon while taking paths hundreds of times more costly than optimal.

2 Related Work

Reward shaping. PBRS characterizes policy-invariant reward transformations [9]. Later work learns when or how strongly to use shaping signals [5, 6], while hindsight methods alter the effective learning signal after failed trajectories [1]. We do not learn a potential: Dijkstra supplies the exact one. Our empirical question is when that exact potential difference improves a fixed finite-budget learner.

Neural algorithmic reasoning. The CLRS benchmark and related neural executors emphasize out-of-distribution size generalization, intermediate supervision, and algorithmic traces [2, 4, 7, 12, 13]. GNARL independently reframes neural algorithmic reasoning through RL

[10]. Our distinction is the controlled use of the same per-decision Dijkstra regret as either a supervised weight or an RL reward on the same evaluation instances.

Learned search and solvers. Differentiable planning [15], learned SAT procedures [11, 14], and learned combinatorial heuristics [8] motivate separating completion from search efficiency. Instruction-pointer executors similarly expose the importance of execution state [3]. We therefore report success, successful-path cost, backward motion, two-cycles, and both stochastic and deterministic decoding.

3 Task and Protocol

3.1 Local graph walking

Each episode samples a connected weighted graph $G = (V, E)$, source s, and target t. At the current node u, the policy observes $K = 8$ neighbor slots. Each slot contains a validity bit, normalized edge weight, visited bit, and target bit. Two global scalars encode normalized rollout progress and clipped graph size. The policy receives neither the global adjacency matrix nor ϕ. Nodes of degree greater than K expose the K cheapest edges with a deterministic weight–node identifier tie break.

Training begins on $n \in [8, 16]$ and normally expands to $n = 32$ after rolling success exceeds 0.9. The default episode horizon is $4n$. Action masks are computed from observed valid flags; invalid-action rate is a hard integrity metric and equals zero in every reported condition.

3.2 One regret, two interfaces

The graded environment reward is

$$r(u, v) = \frac{-w(u, v) + \beta[\phi(u) - \phi(v)]}{w_{\max}} - c, \qquad (2)$$

plus a terminal goal bonus. At $\beta = 1$ and ignoring the shared step cost, Equation (2) is exactly $-\Delta(u, v)/w_{\max}$. The $\beta = 0$ control still pays edge and step costs; "sparse" is retained only as harness terminology. A third reward gives a binary penalty whenever $\phi(v) > \phi(u)$.

The supervised sibling uses teacher forcing with a soft-KL objective whose per-decision weights derive from the same Dijkstra regret. Small and large supervised models are matched to an RL LSTM and Transformer, respectively. The comparison therefore fixes graph generation, local features, regret definition, parameter scale, and approximate model MACs while allowing the training state distribution and optimizer to differ.

3.3 Policies and optimization

RL policies use an MLP, recurrent LSTM, or slot-token Transformer encoder. Q1 compares the LSTM and Transformer with their parameter-matched supervised counterparts. RL uses PPO with Muon at learning rate 0.015 and $\gamma = 0.995$. The potential term in Equation (2) is undiscounted, so we claim the exact Δ identity but not exact discounted PBRS invariance for the PPO objective.

Table 1: Q1 parameter and estimated-compute matching. Ratios are supervised to RL MAC estimates and exclude environment and optimizer work.

Scale	RL params	Sup. params	5M ratio	25M ratio
Small	183,081	185,192	1.010	1.009
Large	531,202	532,152	0.987	0.986

All learned conditions use training seeds 42, 43, and 44. The seed is applied before model construction to Python, NumPy, PyTorch, PufferLib, action sampling, and graph generation. It appears in every checkpoint, curve, manifest, and evaluation filename.

3.4 Frozen evaluation and metrics

Q1 uses three serialized suites of 1,000 graphs each: train scale $n \in [16, 32]$, near OOD $n \in [33, 48]$, and far OOD $n \in [64, 96]$. Both argmax (greedy) and categorical (sampled) decoding are evaluated. Success is target attainment. Optimality gap is

$$\text{gap}(\tau) = \frac{\sum_{(u,v)\in\tau} w(u, v)}{d(s, t)} \qquad (3)$$

and is defined only on successful episodes. Backward rate is the fraction of valid moves with $\phi(v) > \phi(u)$. We also record whether a trajectory contains a two-cycle $u \to v \to u$.

3.5 Predeclared decision gates

Q1 is interesting if either paradigm differs by more than five success points or 0.5 optimality gap in any matched cell, or if train-to-OOD degradation differs. Q2 requires a $\geq$ 15 point graded-minus-β=0 success gap and non-overlapping seed ranges. Q3 requires one architecture to retain $>$ 80% success at least 8$\times$ beyond training size while another collapses, or a ranking reversal. These are practical effect-size gates, not post-hoc p-value thresholds.

4 Q1: Reward Versus Loss Weight

4.1 Matching

Table 1 shows close parameter and model-MAC matching. The supervised steps selected for the 5M and 25M matches are 200 and 1,000 at small scale, and 1,500 and 7,500 at large scale. We report those approximate matches transparently rather than equating MACs with wall-clock or data exposure.

4.2 Matched-compute results

Table 2 contains the headline 25M comparison. Sampled far-OOD success is 90.80% RL versus 12.77% supervised

Table 2: Q1 at the 25M compute match. Entries are three-seed means in the form success (%) / optimality gap / backward rate (%). Optimality gap is conditional on success; lower gap and backward rate are better.

Scale	Learner	Decode	Train scale n=16–32	Near OOD n=33–48	Far OOD n=64–96
Small	RL	greedy	94.13 / 2.18 / 20.02	90.53 / 3.13 / 25.83	79.13 / 5.29 / 33.94
Small	RL	sampled	97.23 / 2.43 / 19.88	95.77 / 3.70 / 25.69	**90.80 / 6.77 / 33.41**
Small	supervised	greedy	18.47 / 4.70 / 42.36	12.23 / 5.67 / 45.16	4.67 / 14.14 / 46.96
Small	supervised	sampled	75.97 / 14.43 / 38.42	53.33 / 19.16 / 42.65	12.77 / 23.07 / 46.19
Large	RL	greedy	87.10 / 1.92 / 20.66	78.93 / 2.52 / 26.71	69.23 / 3.69 / 33.74
Large	RL	sampled	96.37 / 2.52 / 20.21	92.53 / 3.46 / 26.18	**87.57 / 6.04 / 33.63**
Large	supervised	greedy	66.93 / 2.54 / 22.95	22.90 / 10.17 / 42.48	6.27 / 48.15 / 47.19
Large	supervised	sampled	90.90 / 5.77 / 26.30	62.67 / 20.65 / 41.00	31.33 / 58.85 / 46.23

at small scale, a 78.03-point difference. At large scale it is 87.57% versus 31.33%, a 56.23-point difference. Greedy decoding does not reverse the ordering: far-OOD RL reaches 79.13% and 69.23%, while supervision reaches 4.67% and 6.27%.

The divergence grows with graph size. Under sampled decoding, small-model train-to-far success drops 6.43 points for RL but 63.20 for supervision. Large-model drops are 8.80 and 59.57 points. The three-seed ranges do not overlap in either comparison.

Path behavior points in the same direction. In the 25M far-OOD sampled cells, RL backward rates are 33.41% and 33.63%, versus 46.19% and 46.23%. Successful RL trajectories cost 6.77 and 6.04 times optimal, versus 23.07 and 58.85 for supervision. Conditional gaps require care: a deceptively small greedy gap may summarize only a few successful supervised episodes.

4.3 Gate and interpretation

The Q1 gate passes broadly rather than through a selected outlier. Across all matched cells, 24 exceed the strict five-point absolute success threshold, 24 exceed the strict 0.5 optimality-gap threshold, and six train-to-far degradation comparisons have non-overlapping seed ranges. The maximum absolute success difference is 80.60 points (small, 5M, train-scale, greedy); the maximum gap difference is 52.82 (large, 25M, far-OOD, sampled).

The converged 20K-step supervised reference does not repair far-OOD behavior. Its sampled success is 46.30% at small scale and 29.40% at large scale; greedy success is 8.33% and 7.00%. More optimization is therefore not a simple explanation for the matched-checkpoint result.

The controlled interpretation is exposure to state distributions. The supervised learner grades expert-state decisions, whereas RL optimizes on trajectories induced by its own mistakes and recoveries. The experiment is consistent with exposure bias, but does not isolate it from every difference between PPO/Muon and soft-KL training.

Table 3: Q2 main grid at goal bonus 10. Success is the three-seed mean; gap is graded minus β=0.

Start	Horizon	Graded	β=0	Gap
$n = 8$–16	$4n$	99.7	99.3	0.4
$n = 8$–16	$2n$	97.7	96.5	1.2
$n = 8$–16	$1.25n$	95.1	93.1	1.9
$n = 32$–64	$4n$	97.1	89.3	7.7
$n = 32$–64	$2n$	89.1	80.6	8.5
$n = 32$–64	$1.25n$	**83.2**	71.7	**11.5**

5 Q2: A Predeclared Phase Diagram

The main grid trains an LSTM for 5M agent steps in 54 conditions: three rewards (graded $\beta = 1$, $\beta = 0$, binary backward penalty), three horizons ($4n$, $2n$, $1.25n$), two fixed start regimes ($n = 8$–16 and cold-start $n = 32$–64), and three seeds. The complete grid is then repeated with goal bonuses 1 and 0, giving 162 trainings. Each checkpoint is evaluated by sampled decoding on 1,000 shared held-out graphs.

Table 3 shows a coherent regime map. Larger cold-start graphs create most of the separation, and tighter horizons strengthen it. In the strongest cell, seed ranges separate cleanly (82.1–85.2% graded versus 70.3–74.3% control), but the 11.5-point mean gap remains below the declared 15-point threshold. Binary shaping also reaches 83.2%, equal to graded at reported precision; direction matters, but exact regret magnitude is not uniquely necessary.

Removing the terminal bonus does not uncover a hidden effect. The strongest gaps are 12.2 points at bonus 1 and 12.4 at bonus 0. The latter is not reward-free: $\beta = 0$ still pays edge and step costs, and goal termination removes future penalties. All 162 conditions have zero invalid-action rate.

Q2 is therefore an *honest null*. The evidence does not show that shaping is useless; it bounds its effect under the tested optimization settings. Curriculum and a generous horizon make the directional term nearly redundant on small graphs, while harder initial states yield a repeatable sub-threshold advantage. No learning rate, optimizer, seed, or cell was retuned after seeing the gate.

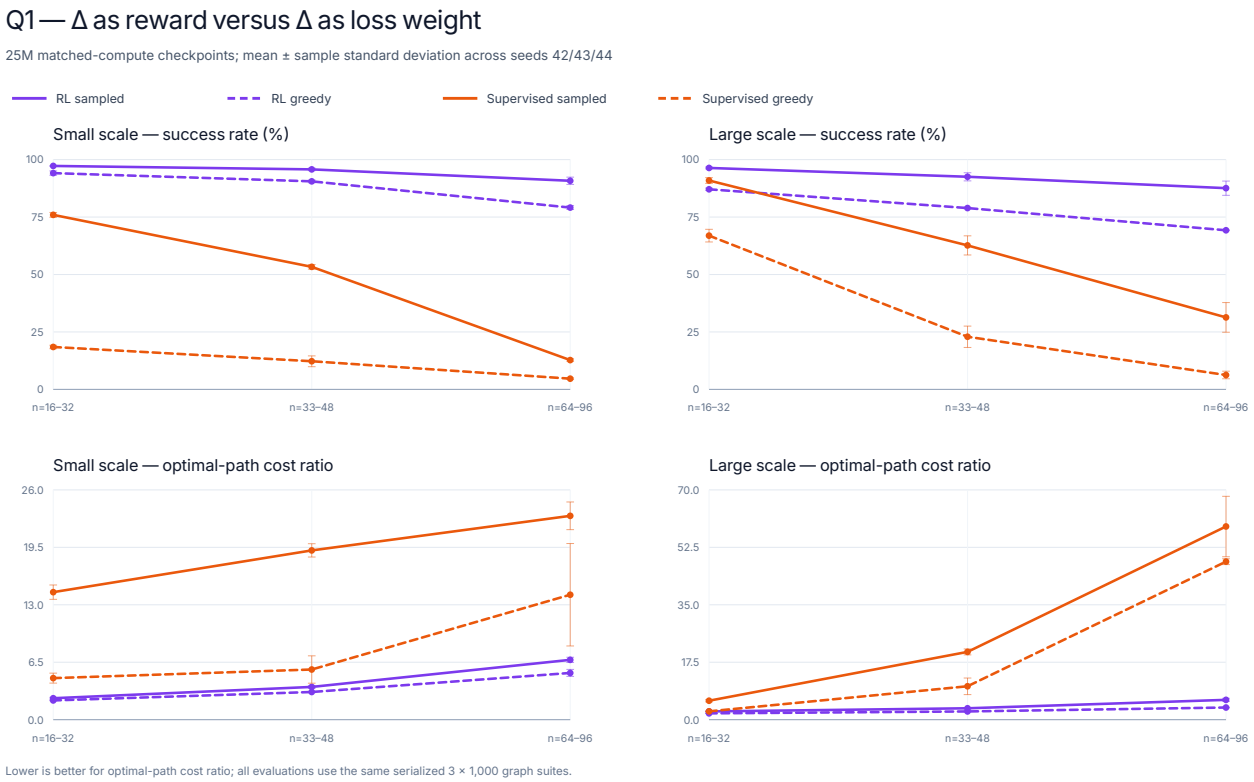


Figure 1: Cross-paradigm Q1 results over all matched parameter scales, budgets, suites, and decoding modes. Error bars are sample standard deviations across training seeds 42, 43, and 44.

Table 4: Sampled-decoding success (%) across graph size. Values are three-seed means; all learned policies train only through $n = 32$.

n	Random	MLP	LSTM	Transf.
32	87.67	96.37	98.30	94.33
48	85.83	94.83	96.73	85.70
96	84.97	93.13	95.83	75.30
192	85.43	91.50	94.50	66.03
384	81.73	90.97	94.00	62.83
768	82.67	**90.93**	**93.57**	56.07

6 Q3: Size Generalization and Decoding

Nine policies—MLP, LSTM, and Transformer at three seeds—train for 5M steps under the curriculum through $n = 32$. Each is evaluated on the same 1,000 graphs per size for $n \in \{32, 48, 96, 192, 384, 768\}$, with a $4n$ budget and both decoding rules. The size feature is clipped at one above the declared $N_{\text{cap}} = 64$, keeping every observation inside the training-time box.

Q3 passes at $n = 384$, twelve times the maximum training size: MLP and LSTM remain above 90%, while the Transformer falls to 62.83%. At $n = 768$, the separation widens to 90.93%, 93.57%, and 56.07%. The result is not uniform Transformer inferiority. On successful $n = 768$ episodes its optimality gap is 22.54, less than half the MLP's 50.11 and the LSTM's 55.32. The Transformer learns a more efficient but brittle policy; the simpler models preserve high-probability reachability with increasingly expensive walks.

6.1 Sampling escapes cycles; it does not solve them

At $n = 768$, greedy success is only 39.27% for MLP, 48.33% for LSTM, and 34.27% for Transformer. Sampled success is higher by 21.8–51.7 points, yet 93.40%, 93.67%, and 92.90% of sampled episodes still contain a two-cycle. Sampling is a stochastic escape mechanism, not evidence that the underlying deterministic controller is cycle-free.

Classical controls make the distinction concrete. Uniform random valid walks retain 82–88% success through $n = 768$ under the generous $4n$ horizon, but their successful-path gap rises from 15.89 at $n = 32$ to 192.27 at $n = 768$ and two-cycle incidence reaches 99.60%. Greedy minimum-edge selection has gap approximately 1 on its rare successes, but success falls from 5.57% to 0.37%. Reachability, efficiency, and local cycling are separate axes.

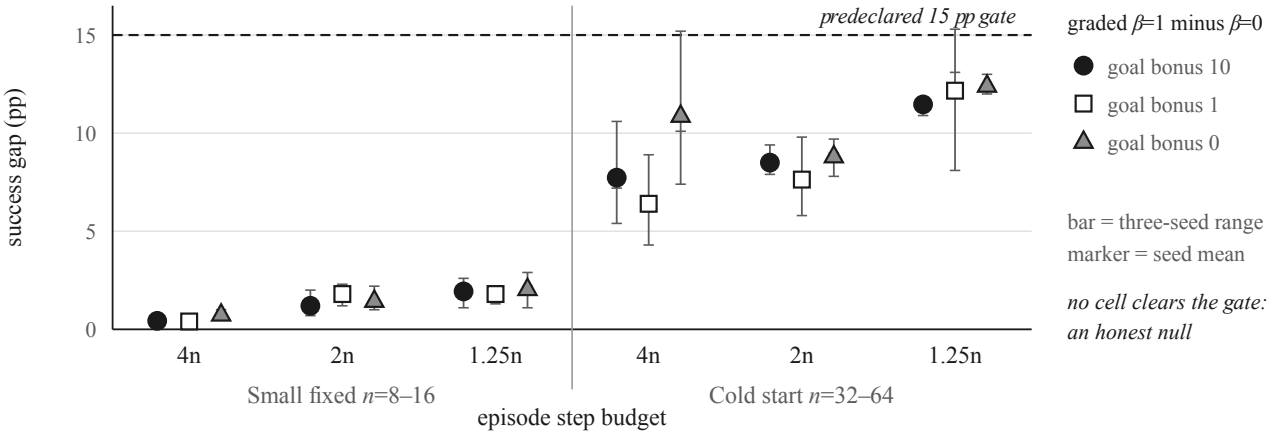


Figure 2: Q2 shaping phase diagram: graded (β=1) minus β=0 sampled success across the six size–horizon cells, at all three goal-bonus levels (162 trainings). Shaping helps systematically in large cold-start, tight-horizon regimes, and the ordering is stable across the bonus ablation, but no cell at any bonus level reaches the predeclared 15-point gate.

7 Discussion

Information and interface are not interchangeable. Q1 controls the graph distribution, local observation, regret definition, parameter scale, and approximate model compute. Its OOD gap suggests that training on self-induced trajectories is valuable when early errors move the policy away from expert states. However, PPO, Muon dynamics, and the number of distinct encountered states remain part of the treatment; the result is not a general causal decomposition of exposure bias.

Policy invariance does not predict learning utility. The identity at $\beta = 1$ is exact, but the usual PBRS theorem does not promise faster PPO learning. Moreover, PPO uses $\gamma = 0.995$ while the implemented potential difference is undiscounted. Q2 is informative precisely because it shows that an exact local direction can become redundant under curriculum, edge costs, goal termination, and a loose horizon.

Sampling is an algorithmic choice. Reporting only argmax or only sampled decoding would yield incompatible stories. Supervised sampling often rescues policies whose greedy rollouts fail, but the rescued paths can be extremely inefficient. In Q3, sampling rescues learned policies from two-cycles without eliminating the cycles. A decoding rule is therefore part of a learned executor, not merely a display setting.

Success is not shortest-path execution. The random-valid baseline rules out interpreting high reachability as Dijkstra-like behavior. Conversely, the min-edge baseline shows that near-optimal conditional path cost can coexist with near-total failure. Claims about learned search require the joint metric vector: success, cost, directionality, cycles, and horizon.

8 Limitations

The graph generator, local feature set, $K = 8$ truncation, and positive edge weights define a specific distribution. The policies do not receive global graph structure, and the comparison does not rank RL and supervised neural algorithmic reasoning in general. Parameter counts and estimated model MACs are closely matched, but wall-clock, optimizer work, and distinct-state counts are not.

Three training seeds support robust practical-effect comparisons but not fine-grained distributional inference. Gate counts are not independent hypothesis tests: cells share training checkpoints and suite construction. Optimality gap is conditional on success and can be unstable when few episodes terminate. The fixed sampling temperature is part of the declared protocol; a wider temperature sweep should be treated as a decoding ablation, not as a newly tuned headline.

The $4n$ horizon permits random exploration to succeed surprisingly often. This is useful as a control but weakens any success-only claim. Stronger algorithmic evidence would require tighter horizons, explicit memory controls, different graph families, or evaluation that directly prices exploration.

9 Conclusion

Exact Dijkstra regret is useful, but how it reaches the learner matters. In this local graph-walking task, RL trained from $-\Delta/w_{\max}$ is markedly more robust than soft-KL teacher forcing on larger held-out graphs at matched model scale and estimated compute. Reward shaping itself produces a coherent but sub-threshold finite-budget advantage once curricula and horizons are controlled. Simple recurrent and feed-forward policies preserve sampled reachability far beyond training size, yet all learned controllers retain severe cycle and path-efficiency failures.

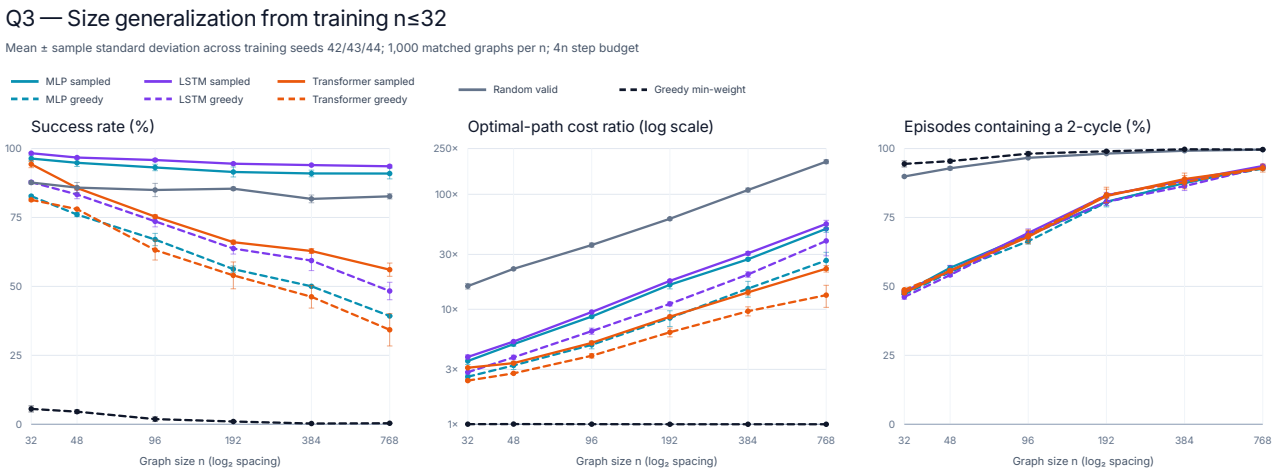


Figure 3: Q3 size generalization. Sampled MLP/LSTM preserve reachability far beyond training size, whereas the Transformer trades success for lower successful-path cost. Greedy decoding exposes a shared cycle pathology.

The strongest supported conclusion is therefore not that one paradigm or architecture executes Dijkstra. Robust learned search is a joint property of the learning interface, induced state distribution, decoding rule, horizon, and path efficiency. Fixed suites, predeclared gates, classical controls, and honest nulls are necessary to see that distinction.

Reproducibility Statement

The merged Q1 dataset contains exactly 72 RL rows and 108 supervised rows; every row has invalid-action rate zero. Q2 contains 162 fixed-suite conditions, also with zero invalid actions. Q3 contains 108 learned evaluation conditions plus classical controls. All checkpoints, curves, manifests, and evaluations encode their training seed. Machine-readable aggregates, complete tables, plotting scripts, and suite manifests accompany the manuscript.

A Complete 5M Q1 Snapshot

Table 5 reports the lower-budget matched comparison. It reproduces the qualitative ordering of the 25M result without selecting a favorable suite or decoding rule.

B Suite and Seed Integrity

The Q1 serialized suites use 1,000 graphs each. Their SHA-256 digests and the graph-generator implementation digest are:

train	`c848ff46125e1cf6998a505a094a6efe`
	`b3e170091a67b7b241623a5a42727463`
near	`dba06cb89c76e674abc8d960ace768ec`
	`a1ac13db2d2111eb724f8a8847b6145c`
far	`d93e4b086a054660f3e40d0e8d7b0170`
	`abb35569bae29e6d8d7a359207d83bbc`
generator	`5a9de1c8ff44c693f797cb09c50d66b2`
	`374a810033b1163767c7ce8f7f3cd9f5`

Hashes were verified before remote evaluation and after local retrieval. Equal-seed 16,384-step checks produce tensor-identical checkpoints and identical non-timing curves; changing the seed changes model initialization.

C Environment Vectorization

The environment precomputes each node's exposed neighbor table at reset and uses NumPy gathers for selected destinations, weights, potential changes, rewards, visited-state updates, and observations. The Python loop remains only for terminated rows, preserving reset and graph-RNG order.

Two 120-step public-output traces, one graded and one binary, remain byte-identical across the scalar and vectorized implementations. Exact graph generation and Dijkstra at reset account for roughly 74% of the remaining full-environment time. We therefore report the 1.30× PPO improvement rather than presenting the 18.72× kernel result as training speed.

D Execution Audit

Q1 ran on a GTX 1080 (compute capability 6.1) using an official CUDA 12.6 PyTorch wheel. At most two GPU processes ran concurrently under reduced priority,

Table 5: Q1 at the 5M compute match. Entries are success (%) / optimality gap / backward rate (%), averaged over three seeds.

Scale	Learner	Decode	Train scale n=16–32	Near OOD n=33–48	Far OOD n=64–96
Small	RL	greedy	90.23 / 2.31 / 20.75	86.10 / 3.41 / 26.28	76.57 / 5.59 / 33.94
Small	RL	sampled	98.13 / 3.05 / 20.43	97.40 / 4.74 / 26.86	95.40 / 8.87 / 33.35
Small	supervised	greedy	9.63 / 3.07 / 44.23	6.50 / 1.35 / 45.69	3.23 / 1.77 / 47.28
Small	supervised	sampled	81.40 / 14.38 / 38.78	72.57 / 21.57 / 41.80	67.77 / 36.78 / 44.14
Large	RL	greedy	87.60 / 1.96 / 20.64	78.73 / 2.42 / 26.66	67.00 / 3.53 / 33.98
Large	RL	sampled	96.00 / 2.42 / 19.91	89.27 / 3.20 / 26.22	77.40 / 4.63 / 33.69
Large	supervised	greedy	34.50 / 3.66 / 35.04	14.20 / 10.13 / 44.84	4.10 / 15.71 / 47.26
Large	supervised	sampled	82.23 / 10.19 / 33.63	59.90 / 20.84 / 42.36	24.50 / 48.62 / 46.22

Table 6: Vectorization benchmarks on 1,024 environments.

Workload	Scalar SPS	Vector SPS	Speedup
Step kernel	267,751	5,013,644	18.72×
Full env.	157,686	316,578	2.01×
PPO, 500K	73,867	96,141	1.30×

with OMP, MKL, and OpenBLAS thread counts capped at one. Training and evaluation logs were retained per seed. Two first attempts failed before model execution because `/usr/bin/time` was unavailable; each traceback was retained and each wrapper was retried exactly once with shell epoch timing. No scientific crash was silently rerun.

The remote suite hashes matched the pinned local values. Artifacts were streamed back and counted before remote deletion: 24 RL files, 54 supervised files, and 12 evaluation JSONs. Bulk remote artifacts and the shared virtual environment were removed after retrieval; raw logs were retained.

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